

HALQA

Investor briefing

The market, the instrument, the architecture, the mechanisms, the economics and the implementation status.

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Sections 3 to 8 are the five functions the platform performs, in the order money moves through them. Section 9 is what those functions are assembled into and sold as. Sections 1, 2 and 13 to 15 are the thirty minute briefing.

Market size and participation

Rs 4tn

Estimated annual flow through committees

37%

Of adults participate, roughly 52 million people

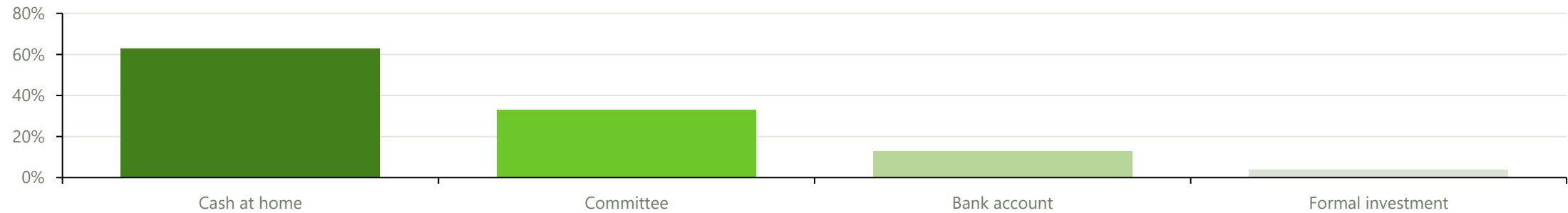
13%

Of adults hold a bank account

63%

Of savers keep their cash at home

The committee is the third largest savings channel in the country. It is not competing with the banking system; it is competing with cash kept at home. Roughly a third of savers already use one.



Where Pakistani savers keep money, per cent of savers. Published national estimates.

Who the member is

The participant is not the destitute household. The academic work on Pakistani committees is consistent that regular income is a precondition of membership, and that groups exclude the irregularly paid themselves. Committees filter on the regularity of income rather than on its level.

MEMBER	MONTHLY INCOME	TYPICAL CONTRIBUTION	WHY THEY QUALIFY
Tailor with a shop	Rs 17,000	Rs 3,000 to 5,000	Daily trade, predictable weekly takings
Driver, salaried	Rs 22,000	Rs 5,000	Salary lands on a fixed day
Site engineer	Rs 45,000	Rs 10,000	Salaried, verifiable employer
Shopkeeper	Rs 60,000	Rs 15,000	Daily takings, long trading history
Daily wage labourer	Rs 18,000	Rs 100 to 250 a day	Income arrives daily, which is why HYPER exists

That distinction is what makes the resulting record a usable credit signal, and it is why the claim is never that this serves the poorest.

The recording gap

A tailor who has paid Rs 8,000 on the first of the month for eleven consecutive months has demonstrated exactly the behaviour a lender spends money trying to establish. That behaviour is recorded nowhere. It exists in a notebook, in a WhatsApp group, or in the memory of the organiser, and when the cycle ends it disappears.

Problem one

The arrangement is fragile. It depends entirely on one person holding the money and on the group's memory of who has paid. There is no written record, no receipt and no recourse.

Problem two

An entire class of credit data is generated at national scale and then thrown away. A household reliable for a decade arrives at a bank as a stranger.

Halqa addresses the second, and the first follows from it.

Definition

Each member pays the same amount every period. Each period the entire pot goes to one member. After one rotation every member has paid the same total and received the same total.

Total money in equals total money out. The arrangement holds nothing at rest, earns nothing for itself, and requires no lender and no capital. The only variable between members is when they collect.

	COLLECTS EARLY	COLLECTS LATE
Receives	The pot in advance	The pot at the end
Economically	An advance repaid in instalments	Saving under a commitment device
Charge	None	None
Risk carried	High until the cycle ends	Falls to zero

The same schedule delivers two opposite financial products at the same time, funded entirely by the members themselves.

Worked example

Twelve members, Rs 10,000 a month, twelve months. The pot is Rs 120,000 and is paid in full to one member each month.

MONTH	COLLECTED	PAID TO	THAT MEMBER HAS PAID IN	THAT MEMBER THEN OWES
1	Rs 120,000	Member 1		Rs 10,000 Rs 110,000
4	Rs 120,000	Member 4		Rs 40,000 Rs 80,000
8	Rs 120,000	Member 8		Rs 80,000 Rs 40,000
12	Rs 120,000	Member 12		Rs 120,000 Nothing

Every member pays Rs 120,000 across the year and receives Rs 120,000 once. Nobody is better off in total. Member 1 has effectively borrowed and repaid at no charge; member 12 has saved under a schedule they could not easily abandon. The amount a member owes the group after collecting is the only figure at risk anywhere in the arrangement, and the next section is about that figure.

PRIOR ATTEMPTS

UBL Komitee Account

United Bank Limited, one of Pakistan's largest banks, markets a deposit product under the committee name. It is the most instructive case in the set because it shows what happens when a large institution takes the word and discards the instrument.

FEATURE	HOW IT WORKS	WHAT IT MEANS
Term	12, 18 or 24 months	Fixed, like a recurring deposit
Contribution	Rs 5,000 to Rs 1,000,000 a month	Wide band, aimed at existing bank customers
Payout	Your own accumulated contributions at the end, plus a bank contribution of one quarter, one half or one full monthly installment	You are paid your own money back with a bonus
Rotation	None	No member ever receives early. The defining feature of a committee is absent
Early exit	Permitted without penalty, forfeits the bonus	No obligation to anybody
Missed months	Forgiven through an installment holiday	No discipline, because there is no counterparty
Custody	The bank, entirely. Member money is a bank liability from day one	Requires a banking licence

The rotation is the product. Remove it and what remains is a savings account with a marketing name. Nobody receives a lump sum early, which is the reason people join committees at all.

PRIOR ATTEMPTS

Pakistan

OPERATOR	PERIOD	REPORTED SCALE	STRUCTURE AND OUTCOME
UBL Komitee Account	Current	Rs 5,000 to Rs 1,000,000 a month, terms of 12 to 24 months	A bank deposit carrying the committee name. Full custody, no rotation between members
Oraan	Founded 2018, USD 3m seed in 2021	Its own research put participation at 41 per cent of Pakistanis and about USD 5bn rotating annually	Platform custody. The operator collects contributions and disburses pots. Strangers pooled and vetted, women weighted
Udhaar Book	Current	A committee register inside a digital ledger app used by retailers	No money moves through the app. The organiser records who has paid. Records an assertion, not a settlement
DigiKhata	Current	A competing ledger app with the same feature	Architecturally identical. Two independent Pakistani apps with real distribution stopped at the same wall
Sidra Humaid committees	Roughly seven to ten years, ending December 2022	About Rs 420 million owed across 117 committees and roughly 200 contributors, no written records	Total concentrated custody. Members from social media wired money to one individual. She announced she could not pay and went into hiding

Two failure modes. Hold the money and carry the licence or the fraud risk, or hold nothing and record only an assertion.

PRIOR ATTEMPTS

Middle East, India and Southeast Asia

OPERATOR	MARKET	REPORTED SCALE	STRUCTURE
Money Fellows	Egypt	8m downloads, over 1m customers, about 350,000 monthly active, USD 1.5bn processed across more than 2m completed circles, about USD 60m raised, profitable in 2025	Platform custody as digital trustee and matchmaker. The largest operator in the category anywhere
Hakbah	Saudi Arabia	More than 1.3m registered users, about 70 per cent young savers, about USD 9m raised	Platform operated savings groups under a central bank sandbox permit
Shriram Chits	India	More than 2.2m subscribers, auction turnover approaching Rs 3,000 crore, 465 branches, 65,000 agents	Statutory foreman under the Chit Funds Act with a full security deposit lodged with the Registrar. Auction ordering
Margadarsi Chits	India, sixty years	Turnover about Rs 14,341 crore, over 6m subscribers, 131 branches	Identical statutory model. The two Indian operators show custody can endure at scale if you are a regulated foreman
KSFE	India, since 1969	More than 680 branches, chit values from Rs 1 lakh to Rs 50 lakh	A state owned non banking finance company as foreman. Sovereign trust replaces private reputation
The Money Club	India, founded 2016	Over 200,000 users, about 17,000 clubs, roughly Rs 40 crore pooled, USD 1.7m raised	Platform intermediated with same day payouts and underwriting by graduated trust
MyPaisaa	India	About USD 980,000 raised, roughly 20,000 installs in two years	India's first fully digital registered chit fund. Shows what full registration costs a young company in speed
Mapan	Indonesia	National scale through community organisers	The organiser is treated as the growth engine, which is the position taken here

PRIOR ATTEMPTS

Africa and the United States

OPERATOR	MARKET AND PERIOD	REPORTED SCALE	STRUCTURE AND OUTCOME
StokFella	South Africa, launched 2016	About 42,000 members by 2025, in a market of roughly 800,000 groups and 11m members holding around R50bn	Group account management with platform held funds, serving existing stokvel groups
Tanda by Yahoo Finance	United States, January to May 2018	About 37,000 installs, never in the top 1,500 of the finance category	Platform managed pooling in groups of five or nine. Closed after four months. Encouraging in the Philippines and Mexico, poor in the United States
eMoneyPool	United States, Phoenix, 2010 to November 2019	Nine years of operation	Platform managed pools reporting payment history to credit bureaux. The operator absorbed defaults itself. The cost of standing behind every member is why the facility here is priced and capped
Esusu	United States, incorporated 31 May 2018	USD 130m Series B led by SoftBank Vision Fund 2 in January 2022, at a USD 1bn valuation	Launched as a savings app, found the domestic market too thin, pivoted to reporting rent payments to all three bureaux. The value was the record, not the pooling
Mission Asset Fund	United States	Groups of six to twelve contributing USD 50 to 200 a month, pots of USD 300 to 2,400	A non profit which formalises peer contributions as documented zero interest loans and reports every payment to all three bureaux

What the evidence establishes

FINDING	EVIDENCE	POSITION TAKEN HERE
Custody decides the outcome, and its effect is binary	Operators which held member money either carried a licence and a statutory deposit, as the Indian foremen do, or failed, as the Rs 420m Pakistani case did	Halqa holds nothing at any stage
The value is the record, and it has been measured	Two American operators abandoned pooling entirely and built businesses on reporting payment behaviour. One reached a USD 1bn valuation on that basis	The record is the asset. The committee is the mechanism which generates it
Distribution is agents, not advertising	Every case which reached scale did so through commissioned organisers, from 65,000 agents in India to community organisers in Indonesia	The host is the acquisition channel and is paid on value, never on recruitment depth
An unpriced guarantee is unbounded	The American operator which absorbed defaults for every member ran nine years and closed	The facility is priced, capped at four levels, and recovered as an advance
Taking the name without the rotation does not work	The bank product removes the early payout, which is the only reason members join	The rotation is preserved exactly as it works informally

Custody

Halqa does not take possession of member money at any stage. Contributions move directly from the paying member's account to the receiving member's account over domestic payment rails. Halqa observes each settlement and writes it to a double entry ledger linked to national identity.

CONSEQUENCE	DETAIL
Commercial	Every large committee fraud recorded in Pakistan required a central pool of members' money under one person's control. No such account exists here, and the protection cannot be weakened under commercial pressure because there is nothing to weaken
Regulatory	No balance is issued, nothing is lent, and no payment system is operated. The path to live money runs through company registration and a merchant services agreement, both commercial steps rather than licences
Pricing	There is no float to fund the business from and no balance to lose, which is why the member price is zero and the revenue comes from services and counterparties
Independent	Researchers at ITU Lahore and the University of Washington, funded by the Gates Foundation and Karandaaz, published the same architecture in 2018 on regulatory grounds

What the platform is and is not

HALQA IS

A record. Every settlement carries a receipt, a timestamp and an identity link, admissible under the Electronic Transactions Ordinance 2002

A treasury. It schedules, prices, collects and settles

A credit record. Payment history becomes a portable score which the member can present outside the platform

A distributor of a licensed manager's savings products, at the second stage

HALQA IS NOT

A holder of member funds at any point in any product

A lender. Members fund one another directly

An issuer of balances, wallets or stored value

An operator of a payment system, an investment manager or a custodian

The distinction is not a policy the company has adopted and could later abandon. It is a consequence of where the money physically goes, which means it holds under commercial pressure and can be demonstrated to a regulator from the ledger rather than argued.

The five functions

	FUNCTION	WHAT IT GOVERNS	WHERE IT APPEARS
1	Onboarding	Who is admitted, what is verified, and what each member is shown. Barriers stay low and the filtering is invisible	Section 4
2	Collection	How money is taken. Collection is attempted on the member's payday, not on the due date	Section 5
3	Pooling products	Where money sits when it is not moving. Halqa holds nothing, so savings route to a licensed manager with a trustee	Section 9
4	Prevention	The principle that recovery machinery is the wrong thing to build. The right thing is an arrangement in which nobody gains by defaulting	Section 7
5	Recovery	What happens when prevention fails, expressed as instruments which can be executed rather than as intentions	Section 8

Two instructions cut across all five. Consumers and savers are separated at the point of joining, because a member who takes the pot early to spend it and a member who waits in order to save carry opposite risks. Filtering is preferred over enforcement wherever both are available.

Registration, ten steps

STEP		WHAT IS CAPTURED	WHY
1	Mobile number	Number, then a six digit code sent to it	The number is the identity on the platform and the channel for every reminder
2	Code verification	One time code confirmed against the number	Proves the number belongs to the member
3	Name and username	Full name as on the identity card	Circles run on real names. It is how members know who they are trusting
4	Location and work	Home address, city, area, occupation, employer, job title, live GPS pin	Area and job are shown to members as a trust signal. The street address and identity number never are
5	Email	Address for receipts and recovery	Records and account recovery
6	Identity card	Thirteen digits, plus a camera capture of the card	Binds every obligation to a government identity
7	Collection account	Rail, account title, account number or IBAN, and the salary day	Auto collection is how every circle collects, so the account is linked before the account exists
8	Password	Minimum eight characters with letters and numbers	Account credential
9	Application PIN	Four to six digits, confirmed twice	A second lock on every open, separate from the password
10	Review and sign	Every field shown back, then the agreement accepted	Consent is recorded against a version and a timestamp

Identity verification

LAYER	MECHANISM	WHAT IT ESTABLISHES	STATUS
Mobile number	Six digit code delivered over the WhatsApp rail, with SMS fallback	The number is controlled by the applicant	Built
Identity card capture	The card is photographed on the device at signup and the number recorded	A document exists and matches the number given	Built
Account title check	At instrument linking the aggregator returns the registered account holder name, which must match the identity card name	The collection account belongs to the member and not to a third party	Built
NADRA Verisys	Identity number checked against the national database through a signed institutional agreement	The identity number is real and belongs to that name	Procurable, deferred past launch on cost
NADRA biometric and facial	Multi biometric verification through the Nishan Pakistan portal, exposed to onboarded institutions since 2025	The living person is the holder of the card. The only check which defeats a borrowed or purchased identity	Paperwork is the long pole, started in parallel
Income verification	Salary slip, or a payday pattern established from the platform's own collection record over two months	Capacity to carry the contribution. The second route needs no document and costs nothing	Built

The verification result and the reference are stored. Raw biometric templates and facial images are never stored.

Facial verification triggers and cost

A facial check on every action would consume money and abandon members at the same time. A NADRA facial certificate has been reported at Rs 20, and one wallet operator charges its users Rs 99 for a check which fails. The check is therefore fired only where it buys something.

TRIGGER	WHY IT MATTERS
Onboarding	Establishes the person once
A new device	The most common route for a stolen credential
A contribution above the configured threshold	Value at risk justifies the cost
Claiming a seat in the first half of the order	These seats carry the exposure
Any exit or hardship request	An impersonated exit is the cheapest attack on the restitution engine
Return after ninety days dormant	The account may have changed hands

The application PIN sits underneath all of it. It is required on every open of the application, on every join commitment, on every exit request and on every payout release, and it is separate from the account password.

Where the handset supports it, the device biometric is layered on the PIN. It never leaves the device and is neither transmitted nor stored.

Affordability and what the member is shown

CAP	LIMIT
Total contributions as a share of net monthly income	33%
Contributions plus known debt service	40%
Total forward exposure, every seat early	4x monthly income
Concurrent circles, income unverified	1
Concurrent circles, income verified	4
Concurrent circles, host with no completed circle	2
Concurrent circles, host after five clean completions	Manual review

The concurrency ceiling on hosts is also the control against one person assembling an artificial network of circles, which is the mechanism behind every large committee fraud.

The 40 per cent figure anchors to the State Bank's cap on the debt burden ratio for consumer financing, reduced from 50 per cent in 2021. Halqa sits stricter on contributions alone.

The caps govern what the discovery screen displays rather than producing a refusal. A member is never shown a circle they cannot afford or a seat they cannot hold, so the system produces no rejections in ordinary use.

Good payment history unlocks seats, additional circles and reduced friction. It never raises the money cap. Only new evidence of income does that, because escalating limits are what produce liquidity defaults.

How one contribution moves

Every rupee that moves through Halqa follows the same eleven steps. Nothing in this sequence passes through an account Halqa controls.

STEP	WHAT HAPPENS	STATE AFTER
1	Schedule written	Scheduled
2	Balance notice	Scheduled
3	Instruction sent	Pending
4	Aggregator routes	Pending
5	Rail settles	Authorised
6	Callback received	Authorised or failed
7	Settlement written	Settled
8	Receipt issued	Settled
9	Round checked	Round complete
10	Payout released	Paid
11	Record furnished	Reported

Why a payment cannot happen twice

A member on a weak signal taps pay, sees nothing happen, and taps again. The phone may have sent both requests. On most systems that is two debits and an apology.

Every write carries an idempotency key: one value naming the exact thing being attempted, stored with the record when it succeeds.

key = member + circle + round + attempt purpose

This is not a refinement added later. Retries are the normal condition on Pakistani mobile networks and under serverless timeouts, so a payment system without it is not partly correct, it is wrong on its most common day.

ATTEMPT	KEY ALREADY STORED	WHAT THE SYSTEM DOES
First tap	No	Charges once, writes the settlement, stores the key
Second tap, same key	Yes	Returns the first receipt. No money moves
Network retry	Yes	Same. The retry is free
Genuine second payment	No, new key	Charges. The member meant it

The same protection covers the payout side. A payout instruction carries the round's key, so a retried release cannot pay a member twice.

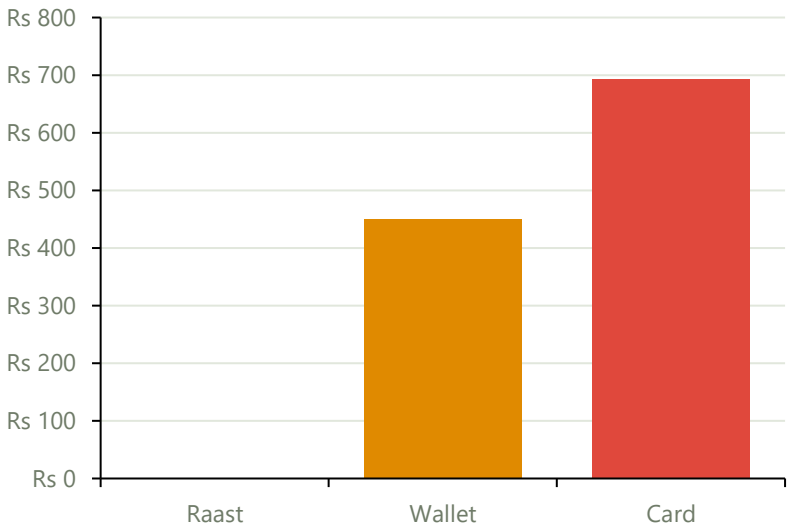
What happens when it fails

FAILURE	CAUSE	RESPONSE	MEMBER SEES
Insufficient balance	Money not in the account yet	Retry ladder runs to the due date. The installment stays open and untouched	A request to pay in one tap
Mandate revoked at the bank	Member cancelled it outside the app	Instrument marked stale, member asked to relink, score untouched for 7 days	Relink prompt
Aggregator timeout	Rail congestion	Reconciliation job queries by idempotency key before any retry, so a silent success is never double charged	Nothing
Partial settlement	Cannot occur	Amounts are integer paisa and the pair must sum to zero, so a half written movement is rejected by the database	Nothing
Payout gate blocked	Another member in the round is unpaid	The cover facility advances the shortfall so the payout releases on time	Paid on time

The design rule across all five: a failure never silently changes what the member owes. The installment either settles or stays exactly as it was.

Rails and cost

RAIL	MEMBER PAYS	HALQA PAYS	POSITION
Raast, the State Bank instant system	Rs 0	About Rs 0	Default rail for every circle type
Wallet to wallet, same network	Rs 0	Rs 0	Where both parties hold the same wallet
Wallet debit under mandate	Rs 0	Rs 300 to 600	Fallback below a Rs 10,000 contribution
Card	Rs 0	Rs 693	Not offered on standard circles
Cash through an agent	Rs 0 to 50	Rs 20 to 60	Last resort. It destroys the digital record



Cost to Halqa of collecting one Rs 20,000 contribution.

Members pay nothing, so every rupee of transaction cost is unrecovered. One twelve member circle running a full cycle on card rails would incur 144 charges at Rs 693, being Rs 99,792 against no member revenue. Raast at approximately zero cost is the only arithmetic under which a product free to members survives, which is why the whole collection design is built on it.

Halqa reaches Raast as a merchant of a licensed aggregator. That is a commercial contract and not a licence.

Collection on payday

The collection event is the member's payday, not the due date, because that is the moment in the month at which the account balance is at its maximum. The due date becomes the deadline for resolving failures.

WHEN	WHAT HAPPENS
Payday eve	A balance check notice goes out by push and over the WhatsApp rail
Payday morning	The mandate executes, before the contribution is even due. Government salaries land on the first, and earlier before Eid. Private payrolls run the first to the seventh
Payday to the due date	On any failure a retry ladder runs alongside a one tap payment request and a message
Due date	Only if every attempt has failed does the member meet the late fee ladder
The 10th, 09:00	A single payout release across the platform

The idea came from a member saying that he pays as soon as his pay arrives. Passive default, where the money arrived and was spent before the due date, is removed entirely. What remains is active evasion, which is a deliberate act taken against a signed consent.

Verifying the declared payday

Pakistan has no open banking regime and Halqa has no bank partner, so the payday is verified from evidence the platform legitimately owns. Four layers, in ascending order of strength.

SIGNAL	MECHANISM	STRENGTH
Account title	At linking the aggregator returns the registered holder name, which must match the identity card	Proves ownership of the account
The platform's own collection record	Every attempt is logged against a calendar day. The first success each month marks a day money was provably present. Two consistent months establish the payday	The strongest of the four, and it costs nothing. If a member declares the wrong day the retry ladder finds the right one
Credit alerts	Optional, member enabled reading of incoming credit notifications. Three fields leave the handset: a day, a coarse amount band, a source class	Two consistent months verify
Salary slip	One photograph, reduced on the device before upload	Documentary verification on day one

A false declaration clears the verification, stops the fee discount, and records a score event of minus 15 with its reason attached, once per window so the same error is never punished twice.

Mandates, receipts and the delinquency ladder

MECHANISM	HOW IT WORKS
Standing mandate	A per circle instruction, on by default at joining with recorded consent. The instrument may be replaced but the mandate cannot be removed while a circle runs. A pause is available with a reason recorded, which costs score rather than access
Mandate verification	Linking an instrument issues a one time code over the WhatsApp rail with a ten minute expiry, and the account title is checked against the identity card name
Instrument storage	Raast ID, wallet or IBAN captured once and masked at rest, maximum five, one preferred. Card linking stores brand, last four digits, expiry, holder and billing address only, never the full number or the security code
Receipts	Every settlement writes a receipt reference and queues a message inside the same transaction as the settlement, so a receipt cannot exist without a settlement or the reverse
Delinquency process	A daily job at 02:00 UTC runs the escalating ladder, server side rather than in the application, so it does not depend on the member opening anything
Grace window	Computed from the period length and never displayed. Score and bureau damage accrue from the first late day regardless of it

Sakh, how the score is computed

Sakh runs from 300 to 850. It is a computed model output, not a running tally, and five inputs produce it.

$$\text{Sakh} = 300 + 550 \times [0.40 \cdot P + 0.25 \cdot U + 0.20 \cdot H + 0.10 \cdot C + 0.05 \cdot M]$$

INPUT		WEIGHT	HOW IT IS MEASURED	WORKED
P	Payment record	40%	Every contribution weighted by age, 0.95 to the power of months elapsed, so a miss last month counts nearly twice a miss a year ago	0.90
U	Liability used	25%	One minus current forward liability divided by the affordability ceiling. Owing Rs 60,000 of a Rs 200,000 ceiling scores 0.70	0.70
H	History	20%	Sixty per cent on circles completed out of five, forty per cent on months on the platform out of twenty four	0.47
C	Concurrency	10%	One minus circles joined in the last ninety days out of three. Three joins in a quarter is distress, not enthusiasm	0.67
M	Product mix	5%	Product types used out of four	0.50

That member scores $300 + 550 \times 0.7213 = 697$, which is the Good band. The plus and minus figures shown in the application are the reason codes explaining what moved the number. They are not the calculation.

What moves the score

EVENT	OPEN CIRCLES	KNOWN CIRCLES	WHY
Contribution paid on time	Plus 2, capped at 40 a cycle	Plus 2	The behaviour the whole model is built to observe
One day late	Minus 15	Minus 10	Matched to the first rung of the fee ladder
Seven days late	Minus 30	Minus 20	Second rung
Fifteen days late	Minus 50	Minus 40	Third rung
Default after collecting	Minus 200	Minus 200	The only event the platform treats as identical across products
Circle completed clean	Plus 200	Plus 200	The atomic unit of proven behaviour
Paying for another member	None	None	Deliberately neutral, so the score cannot be bought
Income newly verified	Raises the ceiling, not the score	Same	Evidence of capacity is not evidence of conduct

Every event is written with its reason attached and is visible to the member on the day it happens. A score a member cannot interrogate is a score they will not trust.

Furnishing to the bureau

A credit bureau does not understand the word committee. It understands tradelines. The translation is straightforward, but one decision inside it determines whether the product delivers what it promises the member.

BUREAU FIELD	WHAT HALQA SENDS	NOTE
Account type	Installment	A seat behaves exactly like an installment obligation
Original amount	Contribution multiplied by the number of rounds	Rs 240,000 on a twelve month circle at Rs 20,000
Installment	The contribution	Rs 20,000
Term	Number of rounds	12
Current balance	Contribution multiplied by rounds remaining	The forward liability, which is the only figure at risk
Opened	The date the member joined	Not the date they collected. This is the decision that matters
Status	Current, 1 to 29, 30 to 59, 60 to 89, 90 plus, written off	Halqa's internal ladder is finer at 1, 7 and 15 days and rolls up to these

Reporting from the join date rather than the collection date means a member who joins in January and collects in November has eleven months of clean reported history before they ever touch the pot. Reporting from collection would discard all of it. That single choice is the difference between a credit product and a payment ledger.

Bureau counterparties

BUREAU	STANDING	POSITION
TASDEEQ, the trading name of Aequitas Information Services Limited, Lahore	The first bureau licensed under the Credit Bureaus Act 2015, with more than 170 member institutions	Primary. Its published score runs from 200 for very poor to 600 for excellent, built from overdues, demographics, loan size, multiple borrowing, credit history length and institution type
DataCheck Limited, Karachi	Founded 2001, the first private sector bureau in the country, also licensed	Second subscriber agreement. Once one furnishing pipeline exists the second costs almost nothing and roughly doubles the number of lenders who can see a Halqa record
eCIB, operated by the State Bank	Accepts data only from licensed financial institutions	Closed to Halqa by design, and nothing in the plan depends on it
Creditinfo Group	Global credit information provider, already working inside the market alongside TASDEEQ and PACRA on scorecards and income estimation models	Not a Pakistani bureau, but the route back into the market if a direct agreement stalls

The commercial position is the strong one. Halqa is not asking for permission to read; it is offering to furnish a population of committee members that no existing subscriber can see. TASDEEQ already counts fintech companies and non bank financiers among its members, so a non bank furnisher is routine rather than exceptional.

Mapping Sakh onto the bureau scale

The two scales differ, so the platform holds both. Sakh is what a member sees and what gates their seat. The bureau equivalent is what a lender outside Halqa reads.

Bureau = 200 + (Sakh – 300) × 400 / 550

Five of the six inputs the bureau scores on are already captured. Overdues come from the delinquency ladder, credit history length from the join date, loan size from the pot, multiple borrowing from the concurrency cap, and institution type from the furnisher registration. The sixth, demographics, the bureau derives itself from the identity number.

The published range should be confirmed with the bureau in writing before any threshold is built against it.

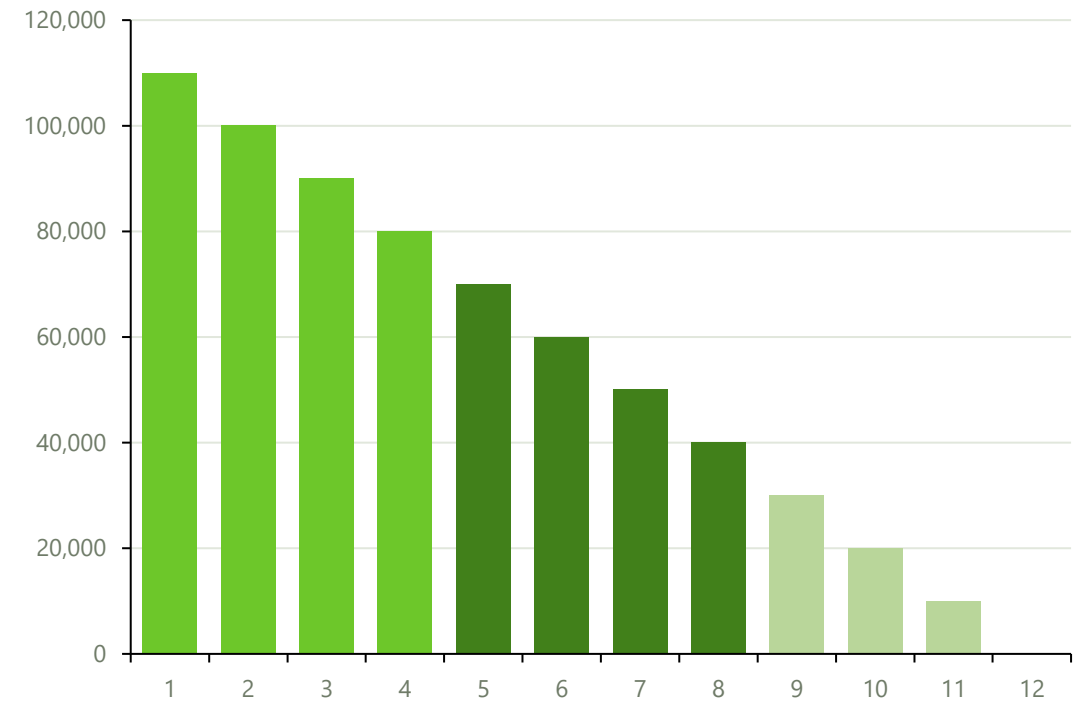
SAKH	BAND	BUREAU EQUIVALENT
300	Rebuilding	200
550	Fair	382
650	Good	455
750	Excellent	527
850	Excellent	600

The quantity at risk

When a member collects, they hold the whole pot and still owe every contribution that remains. That remaining obligation is the only thing at risk anywhere in the arrangement.

It is largest at the first seat, falls by one contribution with each later seat, and is nothing at the last. Risk is not spread across the members; it sits in the early seats and decays.

The platform therefore does not need a system for judging members. It needs a system which decides which member may occupy which seat.



Rupees owed to the circle immediately after collecting, by seat. Twelve members, Rs 10,000 a month.

Seat eligibility

Each score band carries a ceiling on how much a member may owe the group at the moment they collect, expressed in contributions. The ceiling sets the earliest seat they may take.

BAND	MAY OWE AT MOST	EARLIEST SEAT, TWELVE MEMBERS	MARKETPLACE
Rebuilding, below 550	Two contributions	Seat 10	May not purchase
Fair, 550 to 649	Seven contributions	Seat 5	May purchase
Good, 650 to 749	No ceiling	Any seat	May purchase
Excellent, 750 and above	No ceiling	Any seat	May purchase

The same two ceilings apply to a circle of any size, so there is one rule rather than a table per circle size. Above it sits a rule which applies to every new member whatever their score: one of the last three seats only, released after two completed circles with clean records and a manual verification step.

Rs 20,000

The most an unobserved member can cost a circle, on a Rs 120,000 pot

Permanent

A claimed seat never moves. Score gates the seat; it does not reorder anybody

Invisible

Ineligible seats are not displayed, so the rule produces no refusals

No incentive to default

The design does not try to make defaulting difficult for everybody. It ensures that the members carrying the largest obligation are the members who have earned the right to carry it, and that everybody else has nothing to gain by leaving.

POSITION	WHAT THE MEMBER GAINS BY DEFAULTING	WHAT THE MEMBER LOSES
Last seats, where every new member is placed	Nothing. By the time they collect they have already paid almost everything they owe	Platform access, the record they joined for, and 200 points of score
Middle seats	A partial obligation, capped by the band ceiling at seven contributions	The same, plus a recovery case for the outstanding amount, penalties and a 10 per cent rehabilitation fee
Early seats	The largest obligation in the circle	The same, and these seats are only available to members with a proven record they would be destroying
Any seat, asset circle	Possession of an item they do not own	The asset itself, recovered by the party holding title, with proceeds paid into the circle
Any seat, pledged savings	Nothing. The shortfall settles from their own units	The pledged holding, redeemed under an instruction they signed in advance

The seven layers

	LAYER	MECHANISM
1	Identity	Identity card captured at signup, home location pinned, uniqueness enforced on the identity number, the telephone number and the electronic address, so every obligation binds to a government identity
2	Position	Band and tenure define which seats may be claimed, holding worst case exposure to an unobserved member at two contributions
3	Schedule	Fixed dates, escalating reminders, a balance check the day before, and an internal grace window which is never displayed to the member
4	Price	The late fee ladder with matched score damage at every step, and 200 points on a default after collection
5	Payout controls	A payout does not release while any contribution in that round is unpaid; a default on a linked account withholds it; a security shortfall is withheld from the pot up to a ceiling of 60 per cent
6	Consequence	Platform wide feature lock, an internal flag, and a recovery case for the outstanding amount plus penalties plus a 10 per cent rehabilitation fee, identity linked and timestamped
7	Contract	The undertaking, renewed weekly, and the mutual guarantee executed automatically at joining, which runs member to member and never to Halqa

Each layer is independent of the others, so no single failure exposes the circle.

Pricing the cover

The premium is set at roughly one and a half times the default rate the circle type is expected to carry, which builds in a three times margin over expected loss for adverse selection, operating cost and reinsurance.

The relationship does not change with the size of the circle, so one rule prices every product.

Every price in the business is therefore a function of one input which has not yet been measured. The first hundred completed circles establish it and reprice the schedule. That is a stronger position than presenting the figures as fixed, because it says the pricing is a model with a named input rather than a guess.

CIRCLE TYPE	EXPECTED DEFAULT	PREMIUM CHARGED
Known	0.3%	0.5% of each contribution
Open	1.0%	1.5% of each contribution
HYPER	30.0%	15.0% of each contribution

The cover facility is priced, capped per circle, per host, per member and in aggregate against paid up capital, subrogated as an advance rather than a payment, and reinsured above the aggregate cap.

The time value of a turn

Every member pays the same total and receives the same total, so on paper nobody gains. That is true in rupees and false in economics. Rs 240,000 in month one is not the same thing as Rs 240,000 in month twelve, and every member already knows it. That is why they compete for early turns.

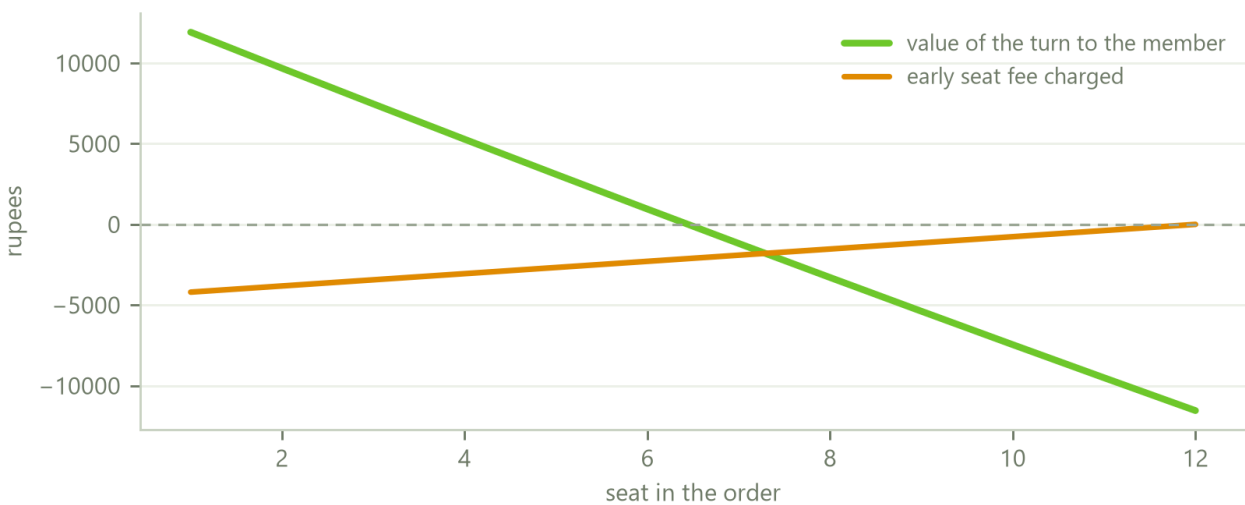
Value of a turn = $\text{pot} \div (1 + i)^k$ – present value of every contribution

Twelve members, Rs 20,000 a month, a Rs 240,000 pot, discounted at twelve per cent a year.

SEAT	POT RECEIVED, PRESENT VALUE	CONTRIBUTIONS, PRESENT VALUE	VALUE OF THE TURN
1		Rs 237,745	Rs 225,829 Plus Rs 11,916
4		Rs 231,077	Rs 225,829 Plus Rs 5,248
6		Rs 226,769	Rs 225,829 Plus Rs 940
7		Rs 224,637	Rs 225,829 Minus Rs 1,192
9		Rs 220,447	Rs 225,829 Minus Rs 5,382
12		Rs 214,286	Rs 225,829 Minus Rs 11,543

The spread between the first seat and the last is Rs 23,459, which is 9.8 per cent of the pot. That value is not created by Halqa and it is not destroyed. It moves from the late members to the early ones, and the only question is whether the arrangement prices it openly or pretends it does not exist.

Where the order turns neutral



Value of the turn against the early seat fee currently charged, by seat.

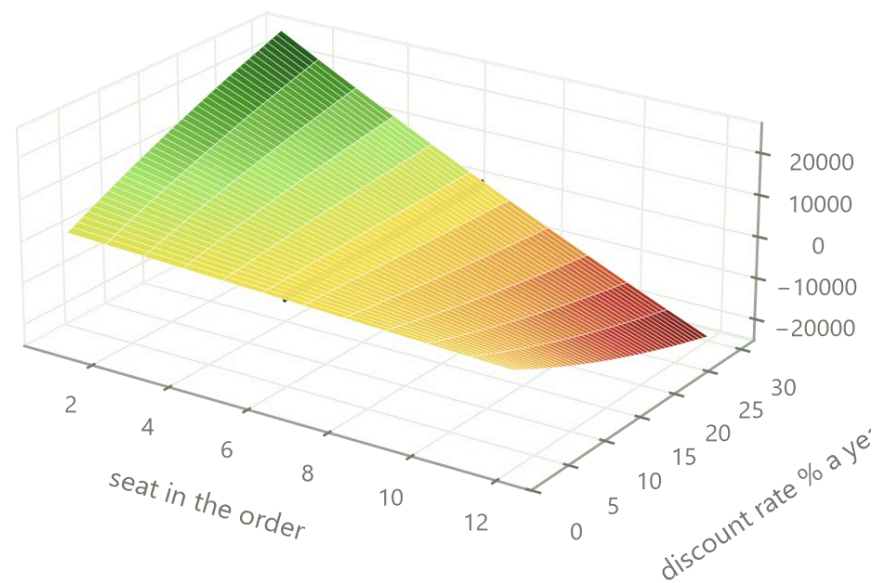
Neutral seat = 6.4 of 12

Setting the value of the turn to zero and solving gives seat 6.4. Seats one to six carry positive value. Seats seven to twelve carry negative value.

The fee schedule does not currently match that. It tapers to zero at the last seat rather than at the neutral seat, and at its steepest it captures Rs 4,200 of the Rs 11,916 the first seat is worth, which is about a third.

The correction is one line: taper the early seat fee to zero at the neutral seat, and credit the seats beyond it from what the early seats paid. The circle then settles its own time value internally and the arrangement is provably fair rather than merely conventionally fair.

Time value across the rate spectrum



Loss Break even Profit

The neutral seat is not fixed. It moves with the rate a member discounts the future at, which in practice is what borrowing costs them.

At two per cent a year the order is nearly flat and the seat barely matters. At thirty per cent, which is close to what a Pakistani household pays on informal credit, the first seat is worth over Rs 25,000 and the last costs about the same.

This is the reason the turn marketplace exists. Members do not value their positions identically, because they do not face the same cost of money. A member with a cash need today and a member saving for a wedding next year sit on opposite sides of this surface, and letting them trade makes them both better off without Halqa taking any risk.

The marketplace commission of ten per cent of the accepted premium is a fee on that trade, not a position in it.

Legal instruments

INSTRUMENT	DETAIL	EFFECT
Undertaking	Electronically signed, renewed weekly, ten clauses, seven day expiry. The typed name must match the registered name exactly. Stored with a hash of the document, the signing IP and a timestamp	A written contract for a fixed sum, which permits recovery by summary suit under Order XXXVII of the Code of Civil Procedure
Mutual guarantee	Executed automatically inside the join transaction. Starting a circle requires every active member to have signed	Runs member to member and expressly not to Halqa, so the circle rather than the platform holds the claim
Consent to bureau reporting	Contained in the undertaking, alongside consent to automatic collection and to the salary anchor	Permits the record to be furnished and a delegated credit check to be run
Service of notice	Consent at signup that legal notice may be served at the pinned address	Recovery does not stall on locating the member, which is the usual failure mode
Evidence pack	Undertaking, guarantee, identity linkage, address, timestamped payment history, circle message log and signature hashes, assembled on demand	Provided to counsel or a court. It is not provided to the circle

Signing a personal guarantee and failing to pay is a civil matter in Pakistan and carries no imprisonment. The only criminal route was the offence relating to dishonoured cheques, and cheques were removed from the model. Recovery requires a court decree and takes months.

Order of recourse

STEP		DETAIL
1	Pledged savings units	Redeemed under the instruction signed at pledge, with the trustee paying the round's recipient directly. Halqa never receives the money
2	Unpledged savings units	Only under a separate signed set off clause, and only after notice with a seven day cure period
3	Pot deductions in live circles	Applied in a fixed order and capped at 60 per cent of the pot, so a member's turn is never emptied
4	Sponsor and restitution claims	Assigned to the cover facility on subrogation
5	Asset circles	The trustee enforces and the sale proceeds are paid into the circle, with any surplus returned to the member
6	Recovery case	Outstanding amount plus penalties plus 10 per cent, enforced by summary suit if pursued

Set off operates in favour of the circle's members and never in favour of Halqa, which is the same construction as the mutual guarantee. Halqa seizes nothing: it instructs a trustee under a pre signed mandate, or hands an evidence pack to a counterparty or a court.

The range

PRODUCT	DESCRIPTION
Known circles	Formed by a host among people who know one another, which is how committees work today. Lighter verification, lower fines, optional insurance, and the group carries the loss
Open circles	Formed among strangers matched by the platform. Heavier verification, steeper late fees, mandatory insurance, and the platform covers a missed contribution from the first round. This product does not exist in the informal market at all, because strangers have no way to trust one another without a record
HYPER committees	Four hundred members, sixty days, daily contributions, pots capped at Rs 15,000. Built for the daily wage earner, who is the largest excluded group and has no savings product today
Asset circles	The pot buys a named item and is paid to the dealer. Possession at the member's turn, legal title held by a security trustee until the obligation is discharged. The only product where the security is a physical thing
Savings layer	Voluntary savings into an Islamic money market fund, held by a trustee in the member's own name. Pledged, it becomes the strongest default control in the model

Known and open circles compared

PARAMETER	OPEN	KNOWN
Joining and contributing	Free, permanently	Free, permanently
Automatic collection	Mandatory	Mandatory
Facial verification	At joining plus the full trigger list	At joining
Manual review by staff	Random sample of 5 per cent	Random sample of 2 per cent
Income detection	A filter. No established payday, no first half seat	A signal. Informs the display, never blocks
Early seat fee, share of pot	1.00 per cent at seat one, tapering to zero	1.75 per cent at seat one, tapering to zero
Insurance premium	1.5 per cent, mandatory, capped at Rs 300	0.5 per cent, optional, capped at Rs 150
Late fee ladder, at 1, 7 and 15 days	3, 7 and 12 per cent	1, 3 and 6 per cent
Score damage at those points	Minus 15, 30 and 50	Minus 10, 20 and 40
Platform covers a missed contribution	Yes, from the first round	No, the group covers
Signed undertaking	Renewed weekly	Signed once at formation
Mutual guarantee	Required	Required
Exit at group approval	No vote, the cover absorbs it	Group vote
Authenticity test	Not applied	Host link confirmed from both ends

Structure

400

Members, fixed. Seven collect on forty days, six on the other twenty

60

Days, fixed. The pot is sixty times the daily contribution

Rs 15,000

Pot ceiling, the level an ordinary daily wage can carry

30%

Default rate the product is priced to withstand

DAILY CONTRIBUTION	POT	COLLECTED EACH DAY ACROSS THE ROSTER		MOST A SINGLE DEFAULT CAN COST
Rs 100		Rs 6,000	Rs 40,000	Rs 5,900
Rs 150		Rs 9,000	Rs 60,000	Rs 8,850
Rs 200		Rs 12,000	Rs 80,000	Rs 11,800
Rs 250		Rs 15,000	Rs 100,000	Rs 14,750

Exposure is bounded by the size of the pot rather than by the size of the roster, which is what makes a thirty per cent default assumption financeable at all. Entry requires two months of demonstrated daily earnings, or a vault holding equal to the pot locked for the sixty days.

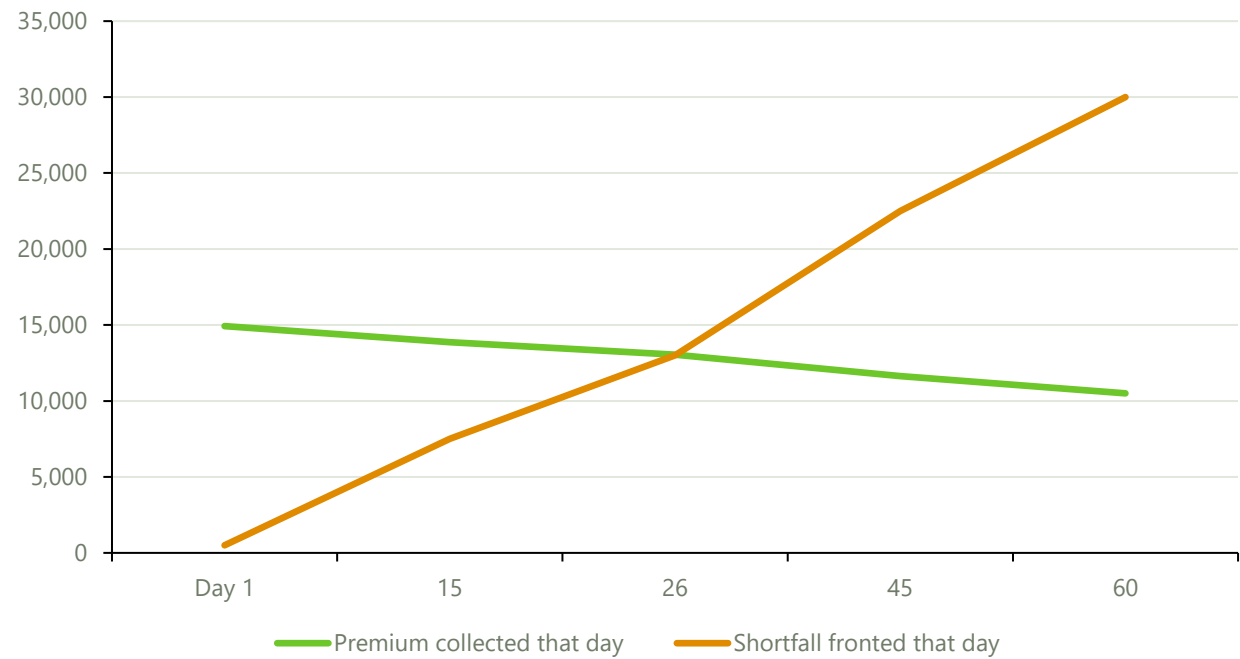
The opening auction

Positions are set once, at the start, by auction, and are fixed for the whole of the cycle. A position may not be sold, bought or altered afterwards, and HYPER committees do not appear on the turn marketplace which operates across the rest of the platform.

STAGE	OPERATION
Bidding window	Twenty four hours, opening when the roster of four hundred completes and before the first contribution falls due
What is bid for	One single named day of the cycle. A member may place as many bids as they wish, but every bid must be for that one day. Bidding across several days is not permitted
What is visible	The standing highest bid on each day, so a member can price a position. No bidder is identified at any point
Allocation	Each day carries six or seven places. The highest bids on that day take them
Everyone else	Members who placed no bid, and members who were outbid, are assigned to the remaining days by a verifiable ballot
Payment	Only a winning bid is charged. An outbid member pays nothing
Eligibility	A bid is refused at the point of placing where the member's band or affordability does not permit that day

Every premium is paid to Halqa in full. There is no member to member premium and no commission split, so the platform receives the whole amount rather than a share of it.

Daily position and funding



Rupees per day, one committee at Rs 250 a day across four hundred members.

LINE	AMOUNT
Deficit at cycle close	Rs 152,250 negative
Auction premiums, taken before the cycle opens	Rs 40,000
Fee uplift of forty five per cent	Rs 50,000
Late fees	Rs 20,000
Residual before recovery	Rs 42,250 negative
Advances outstanding and recoverable	Rs 915,000
Recovery rate at which it closes at zero	4.6%

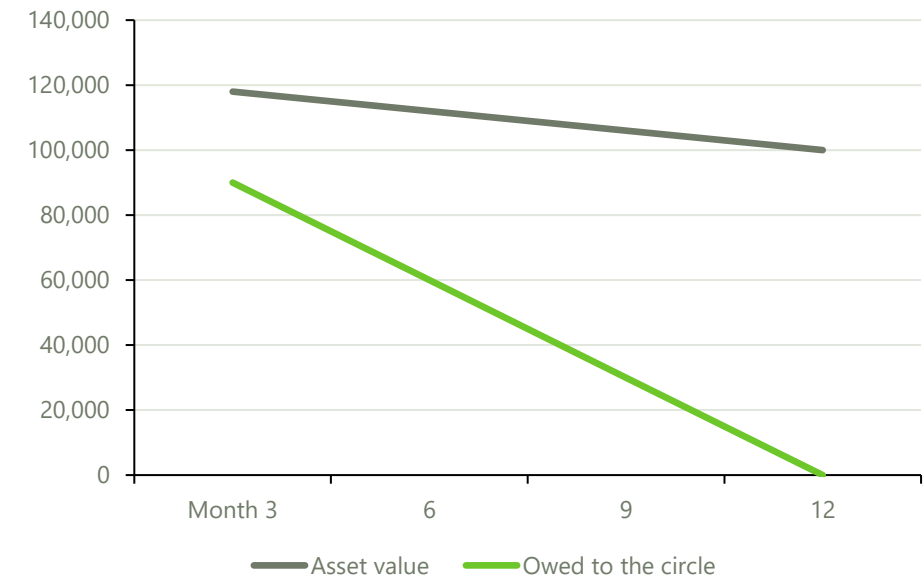
The premium runs ahead of the shortfall for the first twenty six days and behind it after, so the surplus built early funds the deficit late from inside the committee itself. What the product needs is a revolving working balance, not equity.

Retention of title

The pot buys a named item and is paid straight to the dealer. The member takes possession at their turn. Legal title does not pass, and is held by a security trustee for the circle's members until that member's obligation is discharged.

Under the Sale of Goods Act 1930 property passes when the parties intend it to, and a seller may reserve the right of disposal. This is ordinary commercial law rather than an exotic structure.

The obligation falls faster than the asset depreciates, so the security strengthens every month. Every member's contributions end on the same round, so members take possession on different dates and receive title on one date. No member's title depends on any other member's conduct, and if the circle itself fails each retention still stands on its own.



Cover improves from 1.31 times at month three to 3.53 times at month nine.

The savings layer

Committee money earns nothing while it waits, which is a straightforward loss against inflation. Under the 2008 regulations the assets of every regulated fund are held by a trustee and the manager only manages them, so the member's money never touches Halqa.

PRODUCT	MECHANISM	EFFECT
Growth vault	Voluntary savings into an Islamic money market or income fund, units registered in the member's own name	Money which earns nothing today earns a return
Safety vault	The same holding, pledged, with a charge in favour of the circle and an irrevocable redemption instruction signed at pledge	A member cannot default up to the size of the pledged holding. This is what makes short cycle products possible
Float	Contributions collected between payday and the payout date buy units in the payer's own name and redeem on payout day	Small. About Rs 249 a circle a cycle to Halqa. It is a member benefit, not a revenue line

The registration required is a mutual fund distributor registration. It is not an electronic money licence, because no balance is issued, and not a non banking finance licence, because Halqa deploys nothing. One engineering rule is absolute: no payout ever waits on a fund redemption.

One formula, every product

The whole risk model is one expression. Every product below is the same expression with different numbers substituted into it.

Cost of one default

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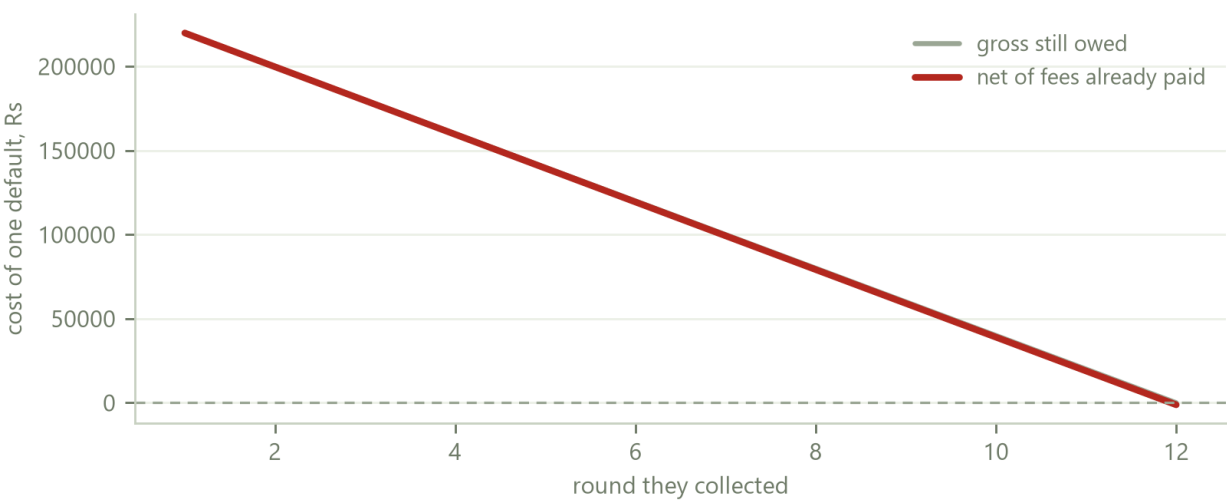
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What the member still owes after collecting, less the fees they have already paid.

Money in equals money out for the member in every column. Halqa's entire take is the fee line.

SYMBOL	MEANING	OPEN MONTHLY	KNOWN MONTHLY	ASSET	HYPER
N	Members	12	12	12	400
T	Rounds	12 months	12 months	12 months	60 days
c	Each installment	Rs 20,000	Rs 20,000	Rs 10,000	Rs 250
φ	Halqa fee per installment	Rs 100	Rs 100	Rs 100	15%, Rs 37.50
P	Pot	Rs 240,000	Rs 240,000	Rs 120,000	Rs 15,000
	Member pays in over the cycle	Rs 240,000	Rs 240,000	Rs 120,000	Rs 15,000
	Member receives	Rs 240,000	Rs 240,000	Asset	Rs 15,000
	Halqa earns per circle	Rs 14,400	Rs 14,400	Rs 14,400	Rs 900,000
	Who covers a default	Halqa	The group	The trustee, then Halqa	Halqa

What one default costs



Cost of a single default on a twelve month circle, by the round they collected.

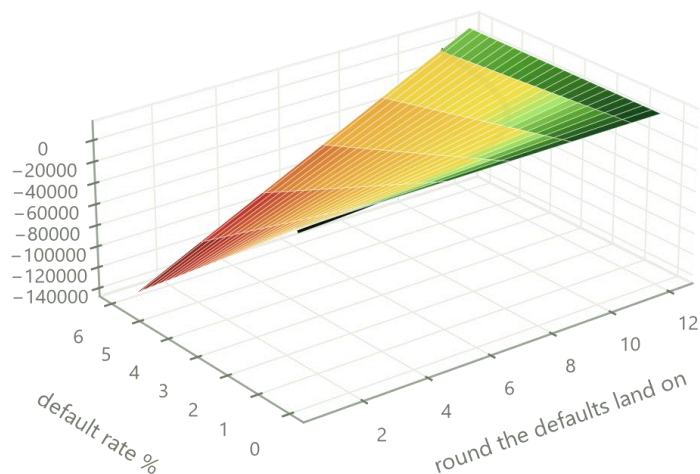
A default in round one costs Rs 219,900, which is fifteen circles' worth of fee revenue. A default in round eleven costs Rs 18,900. Same person, same rules, twelve times the damage.

Break-even round = $cT \div (c + \varphi)$

PRODUCT	BREAK-EVEN POINT	SAFE POSITIONS
Open monthly	Round 11.9	The last one
Known monthly	Round 11.9	The last one
Asset circle	Round 4.0	Nine of twelve
HYPER	Day 52.2	Eight of sixty

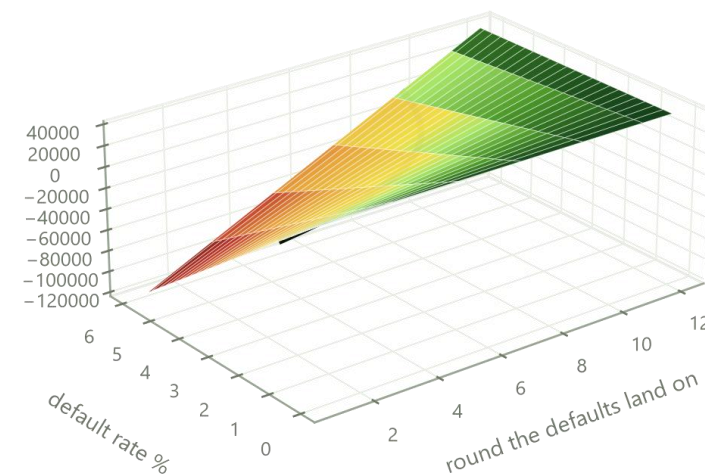
Beyond the break-even point a member who walks has already paid Halqa more than they can cost it. Asset circles reach that point in round four because the trustee can sell the item. Monthly circles only reach it at the very end, which is why position gating carries the whole weight.

Where profit turns to loss, monthly circles



Fee of Rs 100 per installment.

■ Loss ■ Break even ■ Profit



Fee of Rs 275 per installment.

Net position per circle against how many members default and where in the order those defaults land. The black line is break even. At Rs 100 the tolerated default rate is 1.09 per cent against a 1.0 per cent assumption, which is no margin at all. At Rs 275 it is 2.96 per cent, which is roughly three times cover for Rs 175 more a month.

The fee that clears the risk

Break-even default rate

$= \varphi T \div [(c + \varphi)(T - m)]$

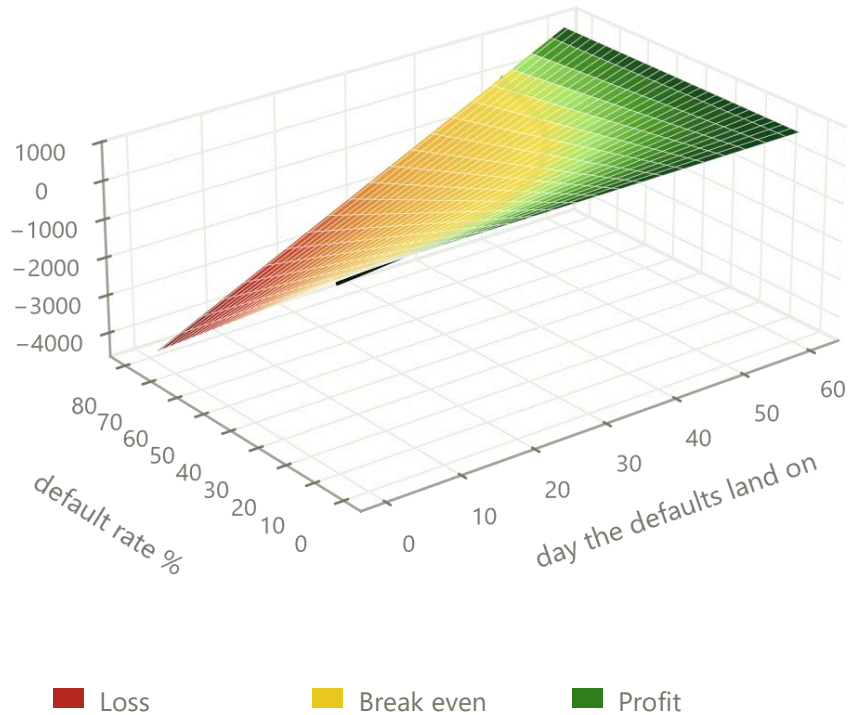
Fee required

$= d c (T - m) \div [T - d (T - m)]$

PRODUCT	DEFAULT RATE ASSUMED	FEE THAT BREAKS EVEN	FEE FOR THREE TIMES COVER	FEE CHARGED	COVER ACHIEVED
Known monthly	0.3%	Rs 28	Rs 83	Rs 100	3.6 times
Open monthly	1.0%	Rs 92	Rs 276	Rs 100	1.1 times
Asset circle	1.0% before recovery	Rs 46	Rs 138	Rs 100	2.2 times
HYPER	30.0%	18% of contribution	not affordable	15%	0.8 times

Two rows are deliberately thin. Open circles at Rs 100 and HYPER at 15 per cent are priced at or slightly below break even, and the shortfall is met by the opening auction, the turn marketplace and late fees. That is a defensible choice, but it should be a stated one. The alternative is Rs 275 on open circles, which the member experiences as Rs 175 a month against Rs 240,000 of money moved.

Where profit turns to loss, HYPER



The same surface for a four hundred member, sixty day committee, run out to an eighty per cent default rate.

The lesson is that the default rate is not what decides the outcome. At the thirty per cent design rate, defaults landing around day ten cost Rs 825,000 and defaults landing around day fifty earn Rs 555,000. Same number of defaulters, a Rs 1.4 million swing, purely on timing.

At thirty per cent the order turns from loss to profit at day 34. Even at eighty per cent, defaults landing after day 51 barely register, because by then the defaulters have paid in almost everything they owe.

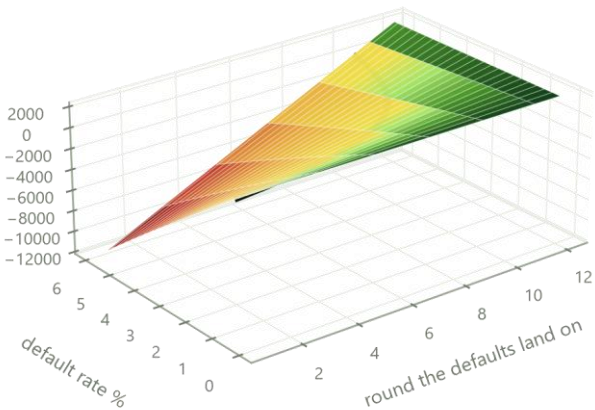
Every control that matters is therefore a control on who occupies the early positions: band gating at the auction, two months of demonstrated daily earnings, and the vault holding.

Who bears the loss

The two monthly products differ in one respect that the profit surface alone conceals: on an open circle Halqa covers the shortfall, and on a known circle the group does.

ONE DEFAULT AT ROUND FOUR, TWELVE MEMBERS, RS 20,000	OPEN CIRCLE	KNOWN CIRCLE
Who fronts the missing contributions	Halqa's cover facility	The other eleven members
Cost to each remaining member		Rs 0
Cost to Halqa	Rs 159,600	Rs 0
What the defaulter gains	Rs 159,600	Rs 159,600
What the member pays in fees	Rs 1,200 a year	Rs 1,200 a year

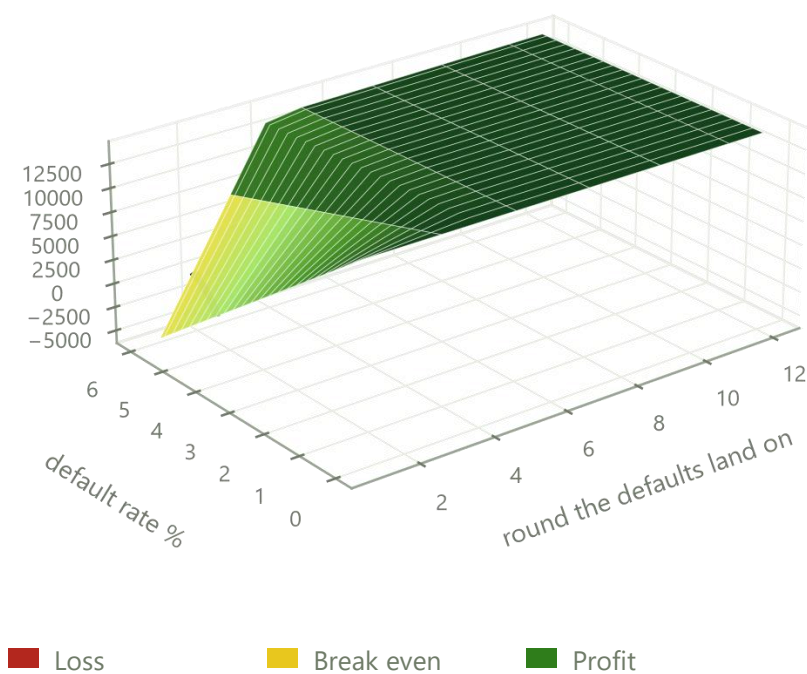
Cost to each member = $c(T - k) \div (N - 1)$
With one flat fee on both products, the open circle member receives cover for nothing while the known circle member pays the same for none. That is the argument for splitting the two prices.



Advantage of a known circle over a covered open circle.

A known circle is cheaper for the member until its default rate passes 1.87 per cent. Beyond that the member is better off paying for cover. Known circles run at 0.3 per cent, so they win comfortably, and the extra fee on an open circle now has an honest name.

Asset circles and the deposit



$$\text{Loss} = \max(0, c(T - k) - \varphi k - h \cdot V(k))$$

ROUND	STILL OWED	ASSET RECOVERS	NET LOSS
1	Rs 110,000	Rs 82,833	Rs 27,167
3	Rs 90,000	Rs 80,500	Rs 9,200
4	Rs 80,000	Rs 79,667	Rs 0
5 and after	falling	rising cover	Rs 0

The trustee holds title, so a default is met by selling the item rather than by pursuing the member. At a seventy per cent forced sale recovery the exposure closes in round four and is zero thereafter.

A deposit removes even that. The threshold is $P(1 - h) \div h$, which is 43 per cent of the pot for complete cover. A twenty per cent deposit halves the exposed window to two rounds and is a realistic ask of a dealer.

REWARDS

What earns a reward

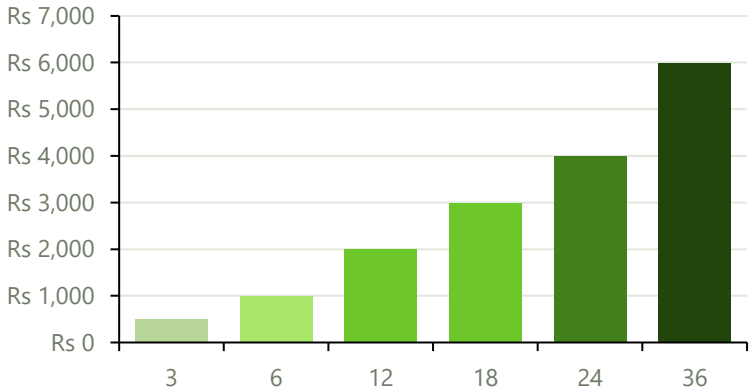
On time payment is the single input on which the whole model rests. Expected loss, the insurance premium, the score and the value of the data asset all move with it, so the reward system is attached to that behaviour and to nothing else.

EVENT	POINTS	SCORE EFFECT	STREAK EFFECT
Contribution paid on or before the due date		10 Plus 2, capped at 40 a cycle	Advances the streak by one
Contribution paid three or more days early		15 Plus 2	Advances the streak by one
Each consecutive clean round	Multiplier of plus 5 per cent, ceiling plus 50 per cent	None	Compounds
Completing a circle with a clean record		200 Existing completion event	Carries the streak into the next circle
Paying a contribution for another member		50 None	Neutral. It does not build a streak, so the ladder cannot be bought
A missed contribution	None	The late fee ladder applies	Resets the streak to zero

Points and score stay separable. Points buy discounts, score gates seats. Points never buy score, and score is never sold.

The ladder

CONSECUTIVE ROUNDS PAID ON TIME	REWARD FROM A PARTNER RETAILER	ALTERNATIVE, A HALQA CHARGE WAIVED
3	5 per cent discount code, capped at Rs 500	None at this tier
6	10 per cent capped at Rs 1,000, plus delivery waived	None at this tier
12	12 per cent capped at Rs 2,000	One month of insurance premium waived
18	Category offer capped at Rs 3,000	Seat placement fee waived on the next circle
24	15 per cent capped at Rs 4,000	Early seat fee waived on the next circle
36	Top tier offer capped at Rs 6,000	Early seat fee and one month of premium waived
Circle completed clean	Completion offer capped at Rs 1,500	200 points



Ceiling on the reward, by streak length.

A member paying Rs 10,000 a month with a clean record for two years reaches the 24 round tier, by which point the ladder has returned roughly Rs 11,500 of retail discount against Rs 240,000 of contributions, none of it paid for by Halqa.

Rewards are never redeemable for cash and release only against contributions which have settled.

Retail partnerships and how they are funded

The discount at every tier is the retailer's own promotional discount, extended to an audience qualified by a year or more of demonstrated payment behaviour. It is customer acquisition spend for them rather than a cost to Halqa, and Halqa earns an affiliate commission on the resulting order at three to twelve per cent of order value depending on category. The reward is the retailer's discount and the revenue is the retailer's commission, so the system funds itself.

PHASE	STEP	WHAT IT REQUIRES
1	Halqa charges waived only	No counterparty. Waivers run through the existing fee engine and can ship immediately
2	Affiliate registration	Registration with the affiliate programmes the large Pakistani marketplaces already operate. A form rather than a negotiation. Links carry an identifier and orders are attributed automatically
3	Direct merchant agreement	A negotiated rate card, discount codes exclusive to Halqa members, placement inside the application, and category exclusivity where obtainable
4	Direct payment	A goal circle's pot is paid to the retailer for a named item and the retailer pays 1 to 3 per cent of the value brokered. This turns the rewards relationship into a distribution channel

Retailers to approach are Daraz, Telemart, iShopping and HomeShopping, the grocery chains Naheed, Alfatah and Imtiaz, and the delivery platforms Foodpanda and Bykea. Halqa receives order attribution rather than basket contents, and purchase behaviour is not sold.

Threat model

The platform holds no member money, which removes the largest prize but not the others. Six attacks are worth defending against, in order of what they would actually be worth to an attacker.

	ATTACK	WHAT THE ATTACKER GAINS	WHAT STOPS IT
1	Impersonated exit or hardship claim	A payout released to the wrong person, or an obligation written off	Facial verification fired on every exit request, plus the application PIN, plus a manual review sample
2	Account takeover by stolen credential	Access to a member's circles and their payout	Refresh token family burn on reuse, facial check on any new device, PIN held in memory only so a reload re asks
3	Synthetic or borrowed identity at signup	A circle seat with no real person behind it	Identity card capture, account title match, NADRA Verisys and biometric, uniqueness enforced on identity number, telephone number and address
4	One person assembling an artificial circle	A pot funded by the platform's cover facility and never repaid	Host concurrency ceiling, the host link confirmed from both ends, and new members restricted to the last three seats
5	Double charge or double payout through a retry	Nothing for an attacker, but a real loss and a real trust failure	Idempotency key on every write, reconciliation by key before any retry
6	Data exfiltration	Identity numbers and payment histories of every member	Parameterised queries throughout, no raw biometrics or card numbers stored, append only audit log, encryption in transit and at rest

Controls in place

LAYER	CONTROL	STATE
Transport	TLS with strict transport security and a content security policy on both applications	Live
Credentials	Passwords hashed with bcrypt at cost twelve. Secret strength asserted at boot, so the service refuses to start weak	Live
Sessions	Short lived access tokens with rotating refresh tokens. Reuse of a refresh token burns the whole family, so a stolen token cannot be replayed	Live
Second factor	Application PIN on every open, every join commitment, every exit request and every payout release, separate from the password and held in memory only	Live
Device biometric	Layered on the PIN where the handset supports it. Never leaves the device, never transmitted, never stored	Live
Rate limiting	Five failed attempts locks the account for fifteen minutes. Thirty in fifteen minutes is blocked globally. Login timing constant so accounts cannot be enumerated	Live
Injection	Every query parameterised through the typed data layer, so string concatenation into SQL is structurally unavailable rather than merely discouraged	Live
Audit	Append only audit log plus a separate security event table. Every consent, signature and administrative action carries an actor and a timestamp	Live
Dependencies	Zero known vulnerabilities on the web package at the last check	Live
Independent test	External penetration test and a formal secure code review	Before public launch

Posture assessed internally at B plus. The gap to A is the two items marked before launch, and they are named here rather than omitted.

Data the platform refuses

Roughly four hundred lending applications were removed from Pakistani app stores for the permissions they demanded rather than for the money they lent. Each item below would improve Halqa's underwriting and each one is refused, because the cost of holding it is delisting, a regulator, or both.

REFUSED	WHAT IT WOULD HAVE GIVEN	WHY IT IS REFUSED
Contact list access	A social graph for collections pressure	The single practice most associated with the removals, and the one the regulator named
Photo gallery access	Documents and proof of assets	No proportionate purpose. Impossible to justify to a store reviewer
Message reading	Bank credit alerts scraped from SMS	Reads every private message to obtain a handful of numbers
Accessibility service screen scraping	Live bank balances without any partnership	Rejected earlier in the design. It is the delisting trap and no balance is worth it
Background location	Movement patterns and workplace confirmation	One precise pin at signup, with consent, is sufficient and defensible
Full card numbers and security codes	Simpler repeat charging	Never persisted. Brand, last four digits, expiry and holder only
Raw biometric templates and facial images	In house matching	Only the verification result and the reference are kept. The template stays with NADRA

The permissions a lender demands are the fastest signal a regulator has for how it intends to collect. Halqa asks for less than a food delivery application.

Engineering invariants

INVARIANT	WHAT IT MEANS	WHY IT EXISTS
All money is integer paisa held as a big integer	No floating point value appears anywhere in the money path. Rates apply as basis points with integer division	Binary floating point cannot represent most decimal fractions exactly, and the residue is invisible until reconciliation fails
Every movement is a double entry pair	Each settlement writes two ledger entries which must sum to zero	The ledger checks itself. A one sided write is detectable rather than silent
Every write carries an idempotency key	A retried request applies once. The key is stored with the record and a duplicate is a no operation	Retries are the normal condition on mobile networks. Without this a member on a weak signal pays twice
Receipts commit with the settlement	The receipt row and the message queue row are written inside the same database transaction as the money movement	A receipt cannot exist without a settlement, and a settlement cannot exist without a receipt
Production migrations are additive only	Hand written SQL that only adds. Nothing is dropped or renamed during a deploy	Old and new code both stay correct while a release rolls out, so there is no window in which the application is broken
Scheduled work runs server side	The delinquency sweep is a job at 02:00 UTC, not something the application triggers	Enforcement cannot depend on a member opening an application

60 unit tests and 352 integration tests pass. The integration suite exercises the money paths end to end rather than mocking them, which is the only way these six can be asserted rather than asserted about.

Platform and data

AREA	DETAIL
Stack	React and TypeScript on the front end, Node and TypeScript on the API, PostgreSQL behind a typed data layer. Both packages type clean
Delivery	Installable on the handset as a progressive web application, with native Android and iOS shells prepared. Urdu and English throughout, with regional greetings
Scheduled work	The delinquency sweep runs server side as a scheduled job at 02:00 UTC rather than in the application, so it does not depend on a member opening anything
Data minimisation	Only the verification result and the reference are stored for identity checks. Card numbers and security codes are never persisted. Location is a single precise pin at signup, never background tracking
Data the platform refuses	Contact list harvesting, gallery access, message reading and accessibility service screen scraping are all rejected, because each one fails either the app store policy or the regulator and is the practice behind the removal of roughly 400 lending applications
Testing	60 unit tests and 352 integration tests, all passing. The integration suite exercises the money paths end to end rather than mocking them

Revenue lines

LINE	BASIS	PAYER
Insurance premium, open circles	1.5 per cent of each contribution, capped at Rs 300	Member, against cover
Insurance premium, HYPER	15 per cent of each contribution	Member, against cover
Early seat fee	1.00 per cent of the pot at the first seat on open circles, tapering to zero at the last	Member collecting early, from the pot
Seat placement fee	Rs 500 flat where Halqa fills a vacant seat, less 20 per cent if income is verified	Member, optional
Turn marketplace commission	10 per cent of the accepted premium	Member, optional
HYPER auction premiums	The whole premium, with no commission split	Member, optional
Late fees	3, 7 and 12 per cent of the contribution by lateness on open circles	Member in default of the schedule
Retail affiliate commission	3 to 12 per cent of order value by category	Retailer
Direct payment commission	1 to 3 per cent of the value brokered	Retailer or dealer
Asset structure origination	2 to 4 per cent of asset value	Dealer or leasing counterparty
Distribution trail on savings	0.5 to 1.5 per cent a year of assets under management	Asset manager
Rewarded video	USD 1.50 to 4.00 per thousand completed views, on one surface only	Advertiser

Members pay nothing to join a circle and nothing to make a contribution. Every line above is either an optional priced service or is paid by a counterparty.

Unit economics

MEASURE	MONTHLY CIRCLE	HYPER COMMITTEE
Structure	12 members, Rs 20,000, 12 months	400 members, Rs 250 a day, 60 days
Pot	Rs 240,000	Rs 15,000
Flow per cycle	Rs 2,880,000	Rs 6,000,000
Insurance premium collected	Rs 43,200	Rs 900,000
Expected loss at the modelled rate	Rs 13,200	Rs 885,000
Underwriting surplus	Rs 30,000	Rs 15,000
Cycles a year	1	6
Where the profit actually is	The underwriting margin	The opening auction premiums

Against Rs 4 trillion of annual flow, a one per cent share carries Rs 40 billion, on which the member facing take alone is between Rs 680 million and Rs 800 million a year, before the retail commission, the direct payment commission and the savings trail. The retail and savings lines compound, because they grow with members rather than with circles.

The two products earn in different places. On a monthly circle the premium comfortably exceeds expected loss and the underwriting margin is the revenue.

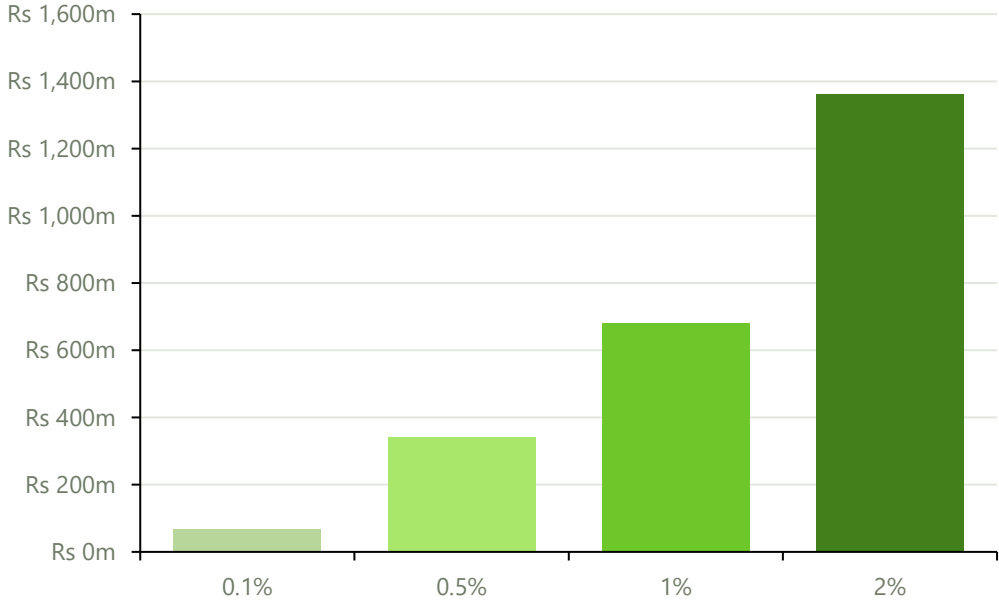
On a HYPER committee the premium is set precisely at the design default rate, so the underwriting line is close to neutral by construction and the revenue is the opening auction, on which the whole of every premium is received rather than a commission on it.

Neither default rate is a Halqa measurement yet. Both are modelled from the literature and from the design point.

Take rate and scale

LINE	SHARE OF FLOW
Insurance premium, open circles	1.50 per cent
Early seat fee, averaged across seats	0.05 per cent
Marketplace and placement fees	0.05 to 0.15 per cent
Late fees	0.10 to 0.30 per cent
Total, member facing	1.70 to 2.00 per cent

The savings trail is the line which does not appear here and which matters most over time. At Rs 1 billion of assets under management a one per cent trail is Rs 10 million a year, and it recurs without further work.



Member facing revenue a year at 1.7 per cent, by share of the Rs 4 trillion market. Modelled, not forecast.

Named counterparties

FUNCTION	CANDIDATE	NOTE
Fund trustee and security trustee	Central Depository Company; Digital Custodian Company	Both qualified. A single sourced trustee is a commercial exposure
Digital asset manager	Mahaana Wealth	Digital first manager for the savings layer
Islamic money market funds	Al Meezan; NBP Fund Management; UBL Fund Managers; Alfalah; HBL; Atlas	Daily dividend funds, built for same day and next day redemption
Leasing counterparty	Orix Modaraba; First Habib Modaraba; Allied Rental Modaraba	Can hold title and write a lease without Halqa holding any licence
Takaful	Pak-Qatar Family Takaful; Salaam Takaful; Jubilee; EFU	Cover above the aggregate cap, and credit life at 0.2 to 0.5 per cent of exposure
Credit bureau	TASDEEQ, being Aequitas Information Services; DataCheck Limited	TASDEEQ is the first bureau licensed under the Credit Bureaus Act 2015, with 170 plus members including fintechs and non bank financiers. DataCheck, founded 2001, is the second agreement
Identity	NADRA, through the Nishan Pakistan portal	Verisys and multi biometric. The longest lead item
Payment rails	1LINK as switch operator; PayFast and Safepay as aggregators	A merchant agreement, not a licence
Messaging	WhatsApp Business Cloud API	Receipts, reminders and one time codes
Retail partners	Daraz; Telemart; iShopping; Naheed; Alfatah; Imtiaz; Foodpanda; Bykea	Affiliate registration is enough for the first phase
Non dilutive funding	Karandaaz Pakistan; Ignite National Technology Fund	Karandaaz co funded the 2018 research validating this architecture
Regulator	SECP regulatory sandbox	The least expensive credibility available

Collection and credit, in production

AREA	WHAT IS RUNNING
Checkout and rails	Gateway style checkout across Raast, JazzCash, Easypaisa, bank transfer and cash, with an explicit failure path which leaves the installment untouched
Saved instruments	Captured once and masked at rest, maximum five, one preferred. Card linking stores brand, last four, expiry, holder and billing address only
Mandate verification	A one time code over the WhatsApp rail with a ten minute expiry, plus the account title check against the identity card name
Automatic collection	A standing per circle mandate, on by default at joining with recorded consent, replaceable but not removable while a circle runs
Payday collection	Balance notice the evening before, execution on the payday morning, a retry ladder to the due date, and four independent verification signals
Sakh	Score from 300 to 850 with bands, credit events with reasons, a movement chart, and an exportable credit record which verifies without a Halqa account
Delinquency	A daily job at 02:00 UTC running the escalating ladder, recovery cases with a 10 per cent rehabilitation fee, and a bureau export queue
Turn marketplace	Listing, bidding and acceptance, premium capped at 50 per cent of the payout, 10 per cent commission, zero premium only on Shariah labelled tiers

The application

Live

In production since 20 July 2026, deliberately not opened to the public

60 / 352

Unit and integration tests, all passing

Rs 0

What a member pays to join or to contribute

SURFACE	STATE
Onboarding	Ten steps: phone verification, identity card capture, live location, occupation, a mandatory collection account and an application PIN
Home	Balance, next installment, eight quick actions, committee cards with live collection progress, recent activity
Committees	Own circles and discovery, invite codes, invitation by WhatsApp, contacts, messages and Skype, a public only filter with its disclosure
Payment and receipts	Every Pakistani rail, card entry which stores nothing sensitive, a processing sequence, and a receipt reference for every settlement
HYPER	Tier selection, the opening auction with a live countdown, standing bids per day and bid entry
Rewards	Streak ladder with unlock states, points balance and the retail partner grid
Account	Score dial, credit report, linked accounts, salary status, recovery cases and settings

The ladder

	RUNG	WHAT IT UNLOCKS	GATE
1	Collection	Settlement rather than assertion. Without it there is no credible record, no data asset and no business	Company registration and a merchant services agreement
2	The record	Verified repayment at volume, which is the raw material for everything above	Circles completed, not registrations
3	The bureau link	The record becomes a formal credit signal and Halqa becomes the supplier of a data class the system does not hold	A contributor agreement with a bureau
4	Commerce	Retailers and institutions pay for access to outcomes. Rewards, direct payment and asset circles sit here	Affiliate registration, then merchant agreements
5	The savings layer	Yield for the member, a trail for the platform, and a pledged buffer which cures default at source	Trustee, asset manager and distributor registration

The near term work divides into engineering, which is under the company's own control, and counterparty work, which carries lead times measured in months and is therefore started in parallel rather than in sequence.

Performance measures

	MEASURE	WHY IT IS THE ONE THAT MATTERS
1	Share of contributions settled digitally	The health of the collection rung, and the single most important figure in the company. A ledger of assertions is worth nothing
2	Circles completed clean	The atomic unit of value. A circle which completes has produced a full record for every member in it
3	Circles per organiser	Whether the growth loop compounds or leaks
4	Measured default across the first hundred completed circles	The point at which the loss rate stops being modelled and becomes measured, and at which every price is recomputed
5	Member to organiser conversion	The only growth loop internal to the product

Registrations, downloads and total value committed can all be raised without the arrangement working at all, and are not tracked as measures of progress.

The market is established. The instrument is a thousand years old. The participants number in the tens of millions.

What has never existed is the record.